Chapter 7
Operationalization of Social-Spatial Determinants of Health

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Introduction

The passage of the Affordable Care Act in 2010 sought to transform US healthcare from a fee-for-service model to a value-based model. This value-based model involves a provider payment system based on patient health outcomes rather than the volume of procedures performed (Porter, 2009). This focus on value-based care models has sparked widespread interest among US health systems to integrate social-spatial determinants of health measures into their health care delivery. Concurrently, there has been growth in electronic health record (EHR) adoption and geocoded datasets, expanding the potential for automated approaches for incorporating these measures into care. In 2014, the National Academy of Medicine (NAM, formerly the Institute of Medicine) recommended the inclusion of standard social-spatial determinants of health measures in EHR systems (Institute of Medicine [IOM], 2014). Nearly 10 years later, there is still no standardization among health systems in their measurement and use of social-spatial determinants of health data. A minimal number of health systems have started to implement and evaluate the use of these data in point-of-care decision-making, population health strategies, and reimbursement models; however, widespread usage is far from the reality among health systems and healthcare providers. This chapter provides a scoping review of how organizations in the US health system are currently using social-spatial determinants of health measures.

The chapter is presented in five sections that each describe a way that social-spatial determinants of health measures are being used by the US. health system: EHR records, risk prediction, resource allocation, reimbursement models, and

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surrogates for individual-level social determinants of health. Each section begins with an overview of how the measures are being used, followed by specific examples of their use. For the specific examples, a description of the social-spatial determinants of health measures used (e.g., specific indices), the rationale for their selection, data sources, and the impact of their use are presented when these data are available in peer-reviewed or gray literature. Examples were selected based on timeliness (i.e., recent publications), completeness of information available, and relevance to the section topic.

Integrating Social-Spatial Determinants into the Electronic Health Record

As of 2019, approximately three-quarters of office-based physicians and nearly all nonfederal acute care hospitals in the USA had adopted a certified EHR system (Office of the National Coordinator for Health Information Technology, 2022). EHRs generally capture demographic (e.g., age, sex, race, ethnicity), behavioral (e.g., tobacco, alcohol, illicit drug use), and clinical (e.g., vital signs, diagnoses, medication orders, procedures, and outpatient, inpatient, emergency department encounters) information in both discrete data fields and free-text clinical notes. Social determinants of health, at the individual or area level, are not typically captured in the EHR. However, over the last decade, there has been increasing pressure on health systems to collect and record these data.

In 2014, the NAM's Committee on the Recommended Social and Behavioral Domains and Measures for Electronic Health Records recommended that The Office of the National Coordinator for Health Information Technology (ONC) and the Centers for Medicare and Medicaid Services (CMS) include the addition of 12 measures to capture 11 social and behavioral domains in the meaningful use regulations for EHRs based on the following criteria: strength of evidence of the association of the domain with health; usefulness of the domain to clinical care decisions, population health policies, and research; availability of a reliable and valid measure; burden of obtaining and/or storing the data; sensitivity to patient discomfort; and accessibility of the data. One of these domains was "neighborhoods and communities compositional characteristics." In this domain, the report suggested that EHRs capture residential addresses and neighborhood median household incomes (IOM, 2014). The committee recommended measuring median household income at the census tract level using American Community Survey (ACS) data. The report justified the use of census tract measures based on the homogeneity of socioeconomic characteristics at this scale and their common use as proxies for neighborhoods. Because the ACS measures only a sample of households each year, the report suggested pooling the data across multiple years to obtain the estimates for income, with the number of years serving as a function of the sample size per year in any given tract (IOM, 2014).

The committee considered several other compositional characteristics for standard inclusion in EHRs: air pollution, allergens, other hazardous exposures, land use, access to nutritious food options, public transportation, parks, health care and social services, education and job opportunities, safety and violence, social cohesion, and social organization. While the committee acknowledged the potential clinical utility of these measures, they noted that the data required to create them are not routinely available in a standardized format; the processes used to create the measures can be complex; a number of different approaches to measurement exist; and the validity and usefulness of different types of measures remain a topic of active research (IOM, 2014). Thus, these domains were not included in the 2014 recommendations (IOM, 2014).

Residential addresses are regularly captured in EHRs for clinical communications and billing purposes. However, nearly a decade after the NAM report, median household income is not routinely incorporated in EHRs. This slow adoption may be the result of a variety of factors including limited resources, lack of incentives or requirements by payors or regulators, and minimal clinician training on how to respond to these data (Wark et al., 2021). Rather, health systems are only recently starting to build and pilot the integration of a variety of social-spatial determinants of health into the EHR. Here, we describe three examples of these efforts.

Cincinnati Children's Hospital Medical Center EHR Alert

In 2015, Cincinnati Children's Hospital Medical Center (CCHMC) built automated alerts in the EHR that triggered a multidisciplinary intervention based on the characteristics of the community in which patients resided (Beck et al., 2019). CCHMC selected two neighborhoods in their service area based on disproportionately high rates of all-cause morbidity and the following social-spatial determinants of health: median household income, housing instability, food insecurity, and poor transportation access. When a child from a high-risk neighborhood is hospitalized at CCHMC, an automatic alert from the EHR is sent to a multidisciplinary team of physicians, nurses, social workers, and community engagement consultants. The EHR alert prompted an in-depth chart review and a bedside huddle focused on the potential preventability of hospitalization, identifiable care gaps (e.g., need for vaccination), and transition needs. When appropriate, patients were connected with additional support during inpatient hospitalization (e.g., social work consultation, school services to help with schoolwork completion in the hospital) and, if needed, transition-related services such as post-discharge nurse visits. In these same communities, CCHMC started to proactively ensure outreach to children with asthma had a ready supply of medications in advance of the fall exacerbation season.

The CCHMC evaluated the impact of these interventions on the neighborhood inpatient bed-day rate (the number of days neighborhood children spent hospitalized divided by the number of children living in the neighborhood). Compared to

before the implementation of these community-targeted programs, CCHMC observed an 18% decrease in the inpatient bed-day rate and a 20% decrease in hospitalizations (Beck et al., 2019). No such reductions were observed in demographically similar neighborhoods. As of 2023, only the combined impact of the bedside huddle and proactive outreach had been evaluated; the impact of these interventions alone has not been published.

The Envirome Web Service

In 2020, the Children's Mercy Kansas City (CMKC) hospital system integrated a clinical decision support tool, the Envirome Web Service (EWS), into care delivery (Kane et al., 2021). The EWS geocoded patient addresses in real time and presented key indicators called envirome indicators to clinicians. The envirome patient profile contained the following indicators, measured at the census tract level: socioeconomic (e.g., median family income, median household income, percent of population in college); demographics (e.g., median age, percent of population under 18, percent of population over 65, population by race); and food access and context (e.g., food desert, urbanicity, low income census tract) (Kane et al., 2021). All these elements were pulled from the Census Bureau and the US Department of Agriculture. A pilot study of the EWS found that 6.1% of clinicians with access to the tool were regular users who opted to maintain the EWS in their custom workflows, logging more than 100 EWS sessions per year (Kane et al., 2021). During the development of the EWS, stakeholders expressed concern regarding a lack of actionable content, potentially explaining this limited adoption. Information on how the EWS was used in care and the impact of the EWS on health outcomes was not available.

Community Vital Signs

With funding from the Patient-Centered Outcomes Research Institute, the Accelerating Data Value Across a National Community Health Center Network (ADVANCE) Clinical Data Research Network (funded by the Patient-Centered Outcomes Research Institute), the Robert Graham Center (RGC), OCHIN (originally called the Oregon Community Health Information Network), and Health Landscape partnered with US federally qualified health centers to develop a program to incorporate a core set of community measures into the EHR, described as Community Vital Signs (Bazemore et al., 2016). This program integrates a novel interface, the Community Vital Signs Geocoding Application Interface (API), which would use residential addresses in the EHR to generate geographic identifiers (e.g., county, census tract). The API links these geographic identifiers to join a core set of Community Vital Signs and returns the linked data to the EHR. The program

included six categories of indicators: built environment (e.g., fast food restaurants and liquor stores per 100,000 population, population density); environmental exposures (e.g., median housing structure age, percent of occupied housing units without complete plumbing facilities); neighborhood economic conditions (e.g., percent of vacant addresses, overall percentile ranking for the Centers for Disease Control and Prevention Social Vulnerability Index (SVI)); neighborhood race/ethnic composition; neighborhood resources (e.g., Modified Retail Food Environment Index, percent of people living more than 1 mile from a supermarket); neighborhood socioeconomic composition (e.g., median household income, percent below 100% the Federal Poverty Level (FPL)); and a social deprivation index (Butler et al., 2013). These indicators were derived from publicly available data sources, such as the US Census Bureau, Environmental Public Health Tracking Network, and the Environmental Protection Agency (Bazemore et al., 2016).

The Community Vital Signs API, like the aforementioned EWS, demonstrated proof of concept that it is technically feasible to geocode patient addresses in the EHR, assign patients a broad range of social-spatial determinants of health measures, and display these measures in the EHR. However, unlike the EHR alert evaluated at CCHMC, the data presented to clinicians by these programs were not designed to trigger a specific clinical intervention. In fact, the creators provide little guidance as to how clinicians and health systems should use the collected social-spatial determinants of health data in their clinical care, limiting the utility of these data in the EHR.

Risk Prediction

One of the potential values of integrating social-spatial determinants of health into the EHR has been the potential for improved risk prediction. Researchers have been exploring whether incorporating social-spatial determinants of health in the predictive modeling of health outcomes can improve model performance for the prediction of a range of outcomes, including hospital readmission, adherence to preventive care, social work referrals, and mortality. Chen et al., (2022) conducted a review of these studies and found that, in most cases, models evaluated the addition of a publicly available deprivation index (e.g., SVI, Social Deprivation Index (SDI), Townsend Index, and Area Deprivation Index (ADI)), measured at the zip code or census tract level, to models containing individual-level demographic information as well as relevant clinical data (e.g., comorbid disease). In most cases, the review found that the addition of social-spatial determinants of health to predictive models did not enhance model performance. However, more recent studies reported that social-spatial determinants of health measures enhanced the performance of predictive models in certain sub-populations. Zhang et al. (2020), for example, evaluated whether the addition of social-spatial determinants of health measures could improve the Simplified HOSPITAL score model prediction of potentially avoidable

30-day hospital readmission or death. The Simplified HOSPITAL score included the following individual-level variables: frequency of prior hospital admissions; urgency admission; last available hemoglobin and sodium levels; discharge from an oncology division; and the index hospital length of stay. The team added the following measures at the census tract level: socioeconomic status (e.g., median income, unemployment rate, percent with high school or high school-equivalent diploma, percent foreign-born, percent without insurance, and percent dual-eligible for Medicaid and Medicare); felony rate; walkability score; Gini income equality coefficient; a composite score reflecting household composition and disability; and a composite score for minority status and language. The study found that the addition of these measures did not improve prediction in the general population but did improve the performance of the tool among individuals on Medicaid, patients 65 years of age or older, and obese patients.

Two studies found that the impact of adding social-spatial determinants of health measures differed by race. Hammond et al. (2020) evaluated how the addition of individual- and area-level social determinants of health impacted risk-adjustment models for annual costs of care, all-cause hospitalization, cardiovascular hospitalization, and death. Models initially included age, sex, the original reason for Medicare eligibility, dual Medicaid enrollment, institutionalization in long-term care, and 83 clinical conditions. Of the seven domains of social determinants of health added to this model, six were individual level (alcohol abuse, access to care, economic status, financial strain, marital status, education), and the seventh was the rural versus urban status of the patient's residence. Without the social determinants of health, these models were sufficient predictors among White patients but underpredicted all-cause hospitalization, cardiovascular hospitalization, and costs among Black and Hispanic patients. Adding the social determinants of health measures improved the performance of the risk-adjustment models in the Black and Hispanic populations but not among the White population. Similarly, Segar et al. (2022) evaluated the impact of adding the Distressed Communities Index, the SVI, and the Graham Center 2011–2015 SDI at the zip code level to a model predicting in-hospital mortality in heart failure. Adding these measures enhanced model performance only among the Black population.

As EHRs begin to bring social-spatial determinants of health data into EHRs through programs such as the EWS and Community Vital Signs, such models could incorporate a broader set of social-spatial determinants that may improve risk prediction, particularly in vulnerable sub-populations of patients. Further evaluation of the health impact of using these predictive models to guide care is needed. This evaluation should continue to explore potential differences in the value of risk prediction by sociodemographic features, as early evidence suggests that the impact of incorporating social-spatial determinants into risk prediction models may vary by patient characteristics.

Health Care Resource Allocation

While the ACA advanced interest in the integration of social-spatial determinants of health into health care, the concept is not new. For decades, the federal government has considered social-spatial determinants of health in resource allocation to support population health. Below, we describe two examples of federal programs that use social-spatial determinants of health measures for resource allocation. The first case involved the 1970 creation of the National Health Service Corps (NHSC) that was created to address shortages in health personnel (Health Resources & Services Administration [HRSA], 2023). The second describes the more recent guidelines for the allocation of COVID-19 vaccines.

Health Professional Shortage Areas

The Health Professional Shortage Area (HPSA) and Medically Underserved Areas (MUAs) or Medically Underserved Populations (MUPs) are designations established by the federal government to identify geographic areas, population groups, or facilities that are experiencing a shortage of healthcare professionals (e.g., primary care, dental care, mental health) (U.S. Department of Health & Human Services [HHS], 2019). The HSPA designation is used by over 30 federal programs to identify areas and populations in need of assistance, including the NHSC, Nurse Corps, Indian Health Services loan repayment program, the CMS HPSA bonus payment program, the CMS rural health clinic program, and the J-1 Visa waiver. Three scoring criteria are applied to classify areas as HPSA: population-to-provider ratio, percent of population below 100% of the FPL, and travel time to the nearest source of care outside of the HPSA (HHS, 2019). MUA and MUP are designated according to an Index of Medical Underservice. This index involves four variables: the ratio of primary care physicians per 1000 population, infant mortality, percentage of the population with incomes below the poverty level, and percentage of the population over 65 years of age or older (HHS, 2019). MUA designations inform grants for community health centers, certification of Rural Health Clinics, and Public Health Service Grant Programs. To be certified as a Federally Qualified Health Center, a clinic must serve an area designated as an MUA or MUP (CMS, 2023).

Vaccine Allocation

In 2020, the National Academies of Science, Engineering, and Medicine (NASEM) was tasked with assisting the Centers for Disease Control and Prevention's (CDC) Advisory Committee on Immunization Practice in developing an equitable allocation framework for COVID-19 vaccines. NASEM recommended that vaccine access

should be prioritized based on the SVI at the US Census tract level or, when data are available, SVI-derived indices that include COVID-19-specific measures (e.g., indicators of comorbidities and health system resource availability) (NASEM, 2020). Schmidt et al. (2021) reviewed the vaccination plans of the 64 jurisdictions to which the CDC dispenses vaccines and found that 43 jurisdictions used a geographic measure of disadvantage, and 37 specifically used a disadvantage index to organize their vaccination distribution process. Of the 37 jurisdictions that used a disadvantage index, 29 used the SVI. Among participating jurisdictions, the reason for the usage of these indices differed. Almost half of the jurisdictions report using the disadvantage indices to allocate more vaccines and vaccine appointments (46%); others used it to identify priority areas and populations (46% of jurisdictions), plan outreach and communications (32% of jurisdictions), plan dispensing sites (22% of jurisdictions), and monitor vaccine receipt (11% of jurisdictions). Additional work is needed to determine whether these methods were successful in an equitable distribution.

Social-Spatial Determinants in Reimbursement Models

Unlike the usage of social-spatial determinants of health in resource allocation in the USA, the use of these determinants in healthcare reimbursement models is relatively new. Many value-based programs do not account for the role that social risk factors have in influencing health outcomes, thus potentially disadvantaging plans or providers that care for patients who, because of social inequities, face disproportionate barriers to health and health care (Breslau et al., 2022). In this section, we will summarize recent efforts in the USA to integrate social-spatial determinants of health measures into healthcare reimbursement models.

In the last decade, there have been efforts at the state and federal levels to adjust reimbursement systems to account for measures of social-spatial determinants of health. As of 2023, there are two state-based Medicaid programs that include measures of social-spatial determinants of health. Since 2017, the Massachusetts Medicaid Agency, MassHealth, has used the Neighborhood Stress Score (NSS7) in their calculation to adjust their system's payments to providers (Breslau et al., 2022). The NSS7 is derived from addresses geocoded at the census block group level using seven census variables that were identified in a principal components analysis of 2013 Massachusetts Medicaid data. These components include the percentage of families with incomes below FPL; percent of families with incomes <200% of the FPL; percent of adults who are unemployed; percent of households receiving public assistance; percent of households with no car; percent of households with children and a single parent; and percent of people aged 25 or older who have no high school degree. While weights were used in the original principal component analysis, the program calculates the NSS7 using an unweighted sum, as the weights varied little across the seven variables (Breslau et al., 2022). The second state-based Medicaid program to incorporate measures of social-spatial

determinants of health is Arizona. In 2021, Arizona's Medicaid Agency, AHCCCS, started adjusting capitation payments to managed care organizations participating in the AHCCCS Complete Care Program. The adjustment incorporates four individual-level social risk factors and a modified version of the SVI at the 5-digit zip code level (Breslau et al., 2022). In 2023, risk adjustment for social-spatial determinants of health will begin at the federal level. CMS implemented a new model called the Accountable Care Organization (ACO) Realizing Equity, Access, and Community Health (REACH) (CMS, 2022). Among the new health equity components of REACH is a health equity benchmark that shifts programming dollars from organizations serving fewer underserved patients to those caring for more, using a risk-adjustment score based on ADIs.

To date, little is known about the impact of these relatively new risk-adjustment programs in the USA. Conversely, such risk adjustments have existed for decades in Europe. Since the 1980s, New Zealand has used an area-level deprivation index to identify geographic areas of need and reallocate funds to the most deprived areas. Initially, New Zealand used a large area estimate of need, the Health and Equity Index of Deprivation, for healthcare fund allocation. However, the scale of the tool overlooked small pockets of need. In 1997, New Zealand's Ministry of Health started to implement the New Zealand Index of Deprivation (NZDep), an index that measured need on the Mesh Block scale (90 people on average) (Huffstetler & Phillips, 2019). The NZDep was created from factors collected in the five-year census of 1991 and comprised nine deprivation characteristics. Deprivation was scaled from 1 (low) to 10 (high). Two main types of deprivation are included in the NZDep: material deprivation (e.g., resources, services, physical environment) and social deprivation (e.g., relationships, roles, and responsibilities of members of society). Areas may have one dominant trait of deprivation or may share a combination of social and material deprivation. The reimbursement adjustment based on the NZDep can result in as much as a threefold increase in healthcare funding (Huffstetler & Phillips, 2019). In the United Kingdom, National Health Services has been adjusting payments using the Carr-Hill formula since 2004. This formula, known as the global sum allocation, adjusts reimbursement on clinic-level characteristics (e.g., age and sex structure) as well as rurality, as determined by population density (Huffstetler & Phillips, 2019). Both the New Zealand and United Kingdom approaches have been found to improve access to care (e.g., shorter wait times); however, data on the impact on health outcomes and health disparities have not been published (Foley, 2018; Huffstetler & Phillips, 2019).

Individual Versus Social-Spatial Determinants of Health

Social determinants of health at the individual level are being integrated into EHR systems, evaluated in predictive models, and used in reimbursement adjustment. In 2015, the Department of Health and Human Services required that health systems use the tenth revision of the International Classification of Diseases codes (ICD-10)

for EHR documentation. This version of ICD codes includes a growing list of codes designed to capture a patient's individual-level social characteristics, referred to as Z-codes (e.g., unemployed, homelessness, and victim of crime) (Truong et al., 2020). The CMS recently instituted a requirement that the health risk assessments used by all Medicare Advantage Special Needs Plans include at least one question from a list of specified screening instruments in each of three domains (food security, housing stability, and transportation access). In its Inpatient Quality Reporting Program, the CMS requires reporting of process measures for the collection of social determinants of health (e.g., percent of patients screened; percent of patients who screen positive). Furthermore, the Joint Commission has issued new standards that require hospitals to screen patients for health-related social needs and provide information about community resources and support services to patients who screen positive.

The collection of individual-level social determinants is often more logistically challenging than obtaining social-spatial determinants of health. Unlike the many spatial measures available from secondary data sources, capturing individual-level data often requires primary data collection that can be disruptive to clinical workflows, perceived as intrusive by patients, and have greater issues with missing data. Truong et al. (2020), for example, reported that the uptake of the ICD-10 Z-codes has been slow and poorly reflects the actual burden of social needs experienced by patients. Thus, social-spatial determinants of health measures are often viewed as a proxy for individual-level social determinants. In some cases, social-spatial measures are viewed as an interim step until individual-level social determinants can be more systematically captured. For example, a Society of General Internal Medicine's Position Statement noted, "Area-level indices using census data to evaluate the social vulnerability of communities are immediately available…such indices can be used until individual social risk data can be collected." A report commissioned by the Department of HHS on area-level social determinants of health stated,

> *Individual-level health-related social needs information is not widely available, and thus, developing measures to directly target funds based on these needs is not currently feasible. As an interim step, area-level measures of social needs or deprivation could be used.*

Viewing spatial-social determinants of health measures as proxies for individual-level measures is potentially problematic. First, using these measures as proxies undervalues the role that community context has in health, independent of individual-level factors. Neighborhoods possess social attributes that could plausibly affect the health of individuals, independent of individual-level factors (Diez Roux & Mair, 2010). Second, the integration of social-spatial determinants of health is important for informing place-level health policies and interventions. Third, although individual- and spatial-level measures of social determinants overlap, such measures are not equivalent (Cottrell et al., 2020). Using social-spatial determinants of health measures in place of individual-level measures may mean that patients who could benefit from targeted interventions are missed (Cottrell et al., 2020).

Conclusions

Healthcare policymakers, payors, and providers are starting to explore the integration of social-spatial determinants of health measures into health care. These determinants are being introduced in EHR documentation, risk prediction, reimbursement models, and allocation of resources for health. Pilot work has demonstrated the technical feasibility of integrating these data; however, adoption has been slow. The delay in more widespread adoption is likely due to be a combination of factors, including limited resources; lack of incentives; and a dearth of guidance on how providers and health systems can respond to this data.

Efforts to bring social-spatial determinants of health measures to health care have been dominated using existing socioeconomic indices (e.g., ADI, SVI) that are based on publicly available data (e.g., U.S. Census Bureau data) and applied to predefined administrative spatial boundaries (e.g., census tract, zip code). Selection of measures is based on logistical demands (e.g., data availability) and evidence of the relation of these measures to health outcomes. However, even the integration of publicly available and well-established indices requires technical, financial, and personnel resources that may present barriers to adoption (Wood et al., 2021). Of concern is that health systems serving populations that would most benefit from using social-spatial determinants of health measures may be less equipped to integrate these measures. Financial support or incentives for capturing these data may help health systems overcome these barriers to adoption.

Early evidence suggests the value of integrating social-spatial determinants of health in risk models differs by sociodemographic and clinical characteristics of the population (e.g., age, race, body mass index). Consistent with these findings, studies have demonstrated a synergistic effect of social-spatial determinants of health, race, and health outcomes, such that stronger associations between these determinants and health outcomes have been observed in Black versus non-Hispanic White populations (Mode et al., 2016). Similarly, stronger associations have been observed in older versus younger populations (Wang et al., 2020). Moreover, there is good evidence that different indices used to measure social-spatial determinants of health are not equally effective in uncovering health inequalities for different population subgroups (Allik et al., 2020). Thus, the measures selected to capture social-spatial determinants of health and the application of these measures in health care should be evaluated in subgroups of the population, as the value of integrating these determinants may differ across groups.

Though theory points to the immense benefit of knowledge of the community context that can have on health care delivery and health outcomes, there is little guidance for providers regarding how to use these data in their practice. Moreover, there is limited research on how responding to these data impacts health outcomes and mitigates disparities. Thus, providers express reservations about using social-spatial determinants of health in the clinical care setting, as they do not see these data as actionable. Funding should be allocated to support such research and curriculum for physicians and allied health professionals should educate students on how to care for patients in the context of their social-spatial health determinants.

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